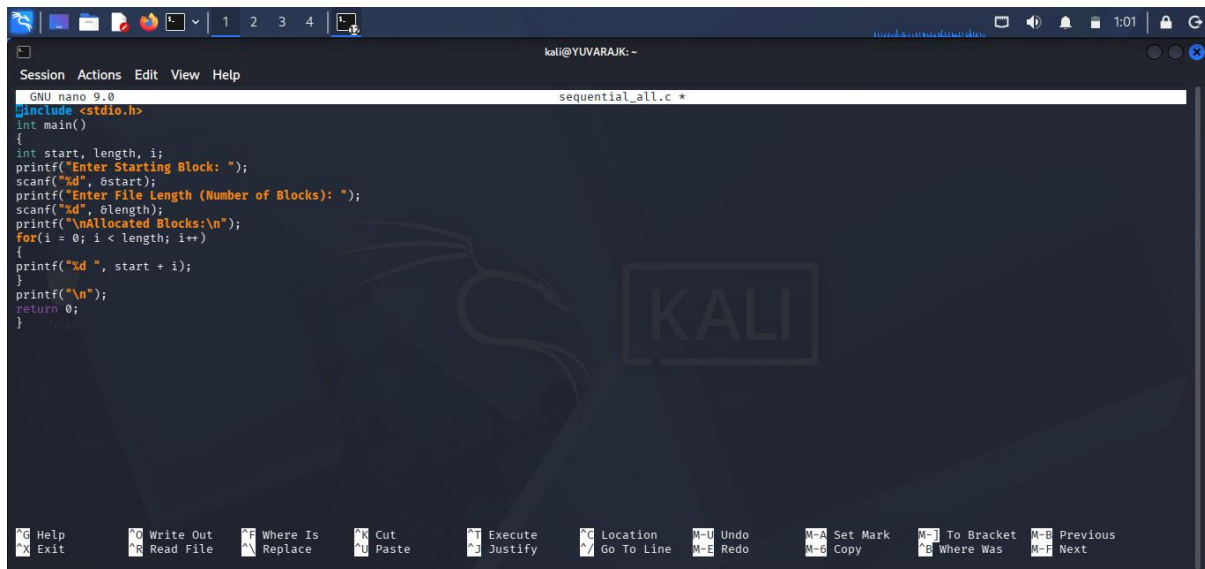
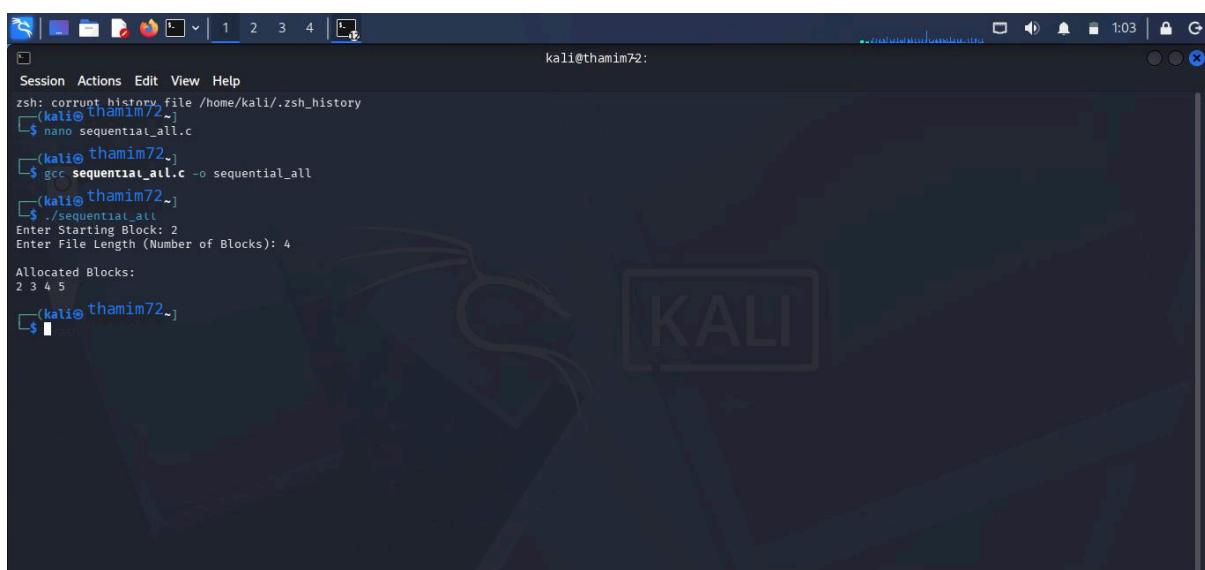


C PROGRAMMING:SEQUENTIAL ALLOCATION



```
GNU nano 9.0 sequential_all.c *
#include <stdio.h>
int main()
{
    int start, length, i;
    printf("Enter Starting Block: ");
    scanf("%d", &start);
    printf("Enter File Length (Number of Blocks): ");
    scanf("%d", &length);
    printf("\nAllocated Blocks:\n");
    for(i = 0; i < length; i++)
    {
        printf("%d ", start + i);
    }
    printf("\n");
    return 0;
}
```

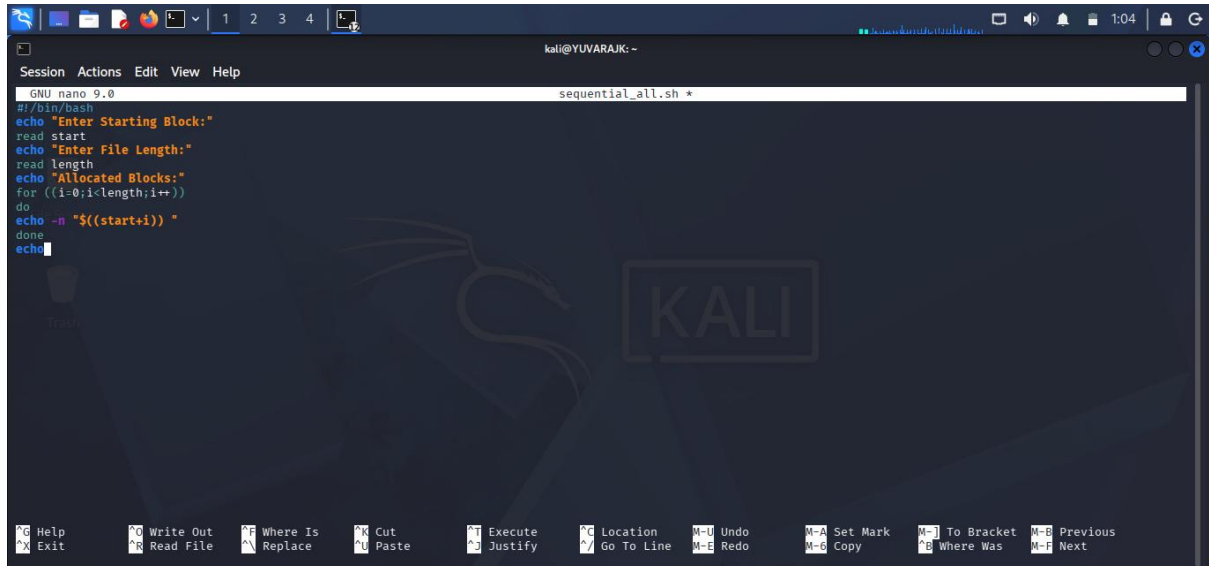
OUTPUT :



```
kali@thamim72:~
zsh: current history file /home/kali/.zsh_history
(kali@thamim72~)
$ nano sequential_all.c
(kali@thamim72~)
$ gcc sequential_all.c -o sequential_all
(kali@thamim72~)
$ ./sequential_all
Enter Starting Block: 2
Enter File Length (Number of Blocks): 4

Allocated Blocks:
2 3 4 5
(kali@thamim72~)
$
```

SHELL PROGRAMMING :SEQUENTIAL ALLOCATION

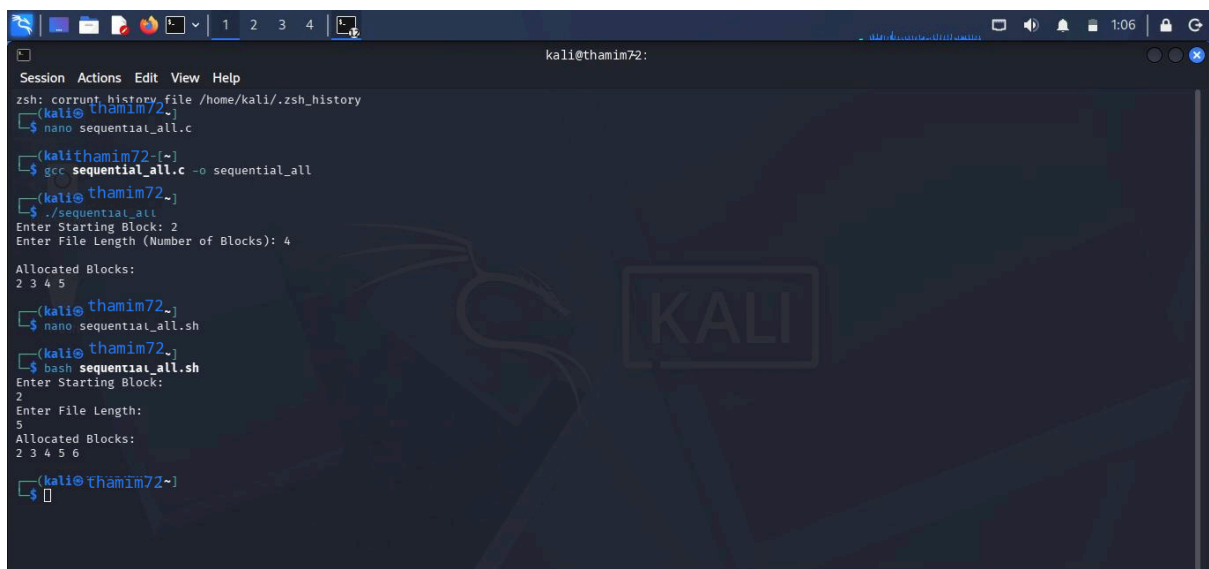


A screenshot of a terminal window with the nano 9.0 editor open. The editor is editing a file named 'sequential_all.sh'. The script content is as follows:

```
#!/bin/bash
echo "Enter Starting Block:"
read start
echo "Enter File Length:"
read length
echo "Allocated Blocks:"
for ((i=0;i<length;i++))
do
echo -n "${start+i}) "
done
echo
```

The terminal window title is 'kali@YUVARAJK: -'. The nano editor's status bar at the bottom shows various keyboard shortcuts like 'H Help', 'X Exit', 'O Write Out', etc.

OUTPUT :



A screenshot of a terminal window showing the execution of the script. The user is at the 'kali@thamim72' prompt. The terminal history shows the following commands and output:

```
zsh: current history file /home/kali/.zsh_history
kali@thamim72~$ nano sequential_all.c
kali@thamim72~$ gcc sequential_all.c -o sequential_all
kali@thamim72~$ ./sequential_all
Enter Starting Block: 2
Enter File Length (Number of Blocks): 4
Allocated Blocks:
2 3 4 5
kali@thamim72~$ nano sequential_all.sh
kali@thamim72~$ bash sequential_all.sh
Enter Starting Block:
2
Enter File Length:
5
Allocated Blocks:
2 3 4 5 6
kali@thamim72~$
```

The terminal window title is 'kali@thamim72:'. The background features a Kali Linux logo watermark.

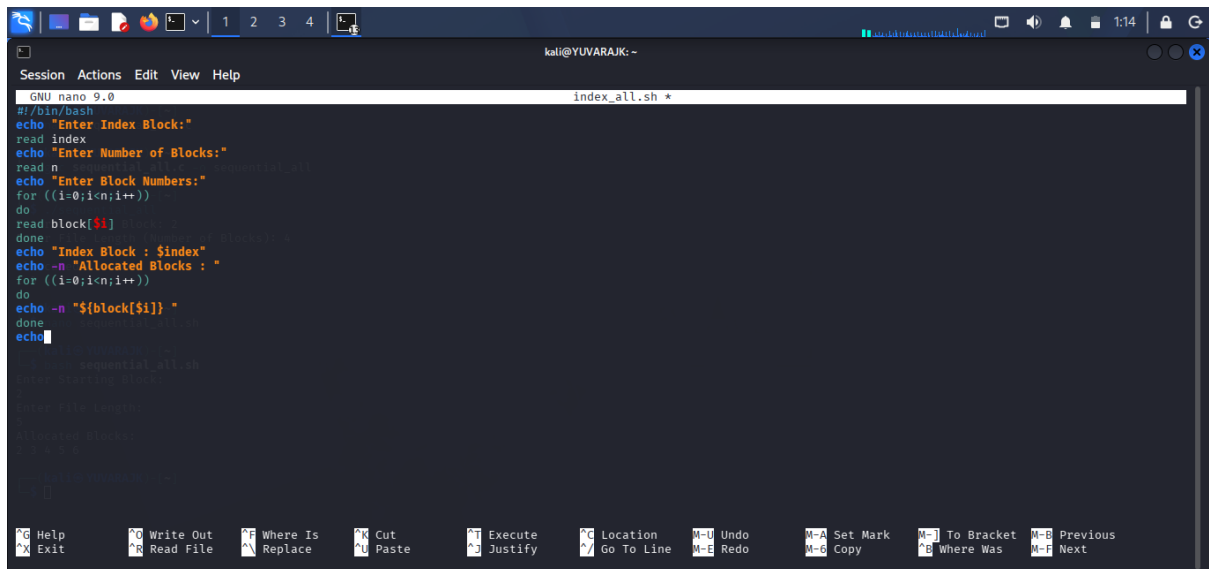
C PROGRAMMING : INDEXED ALLOCATION

```
GNU nano 9.0 index_all.c *
#include <stdio.h>
int main()
{
    int n, indexBlock, blocks[20], i;
    printf("Enter Index Block: ");
    scanf("%d", &indexBlock);
    printf("Enter Number of Blocks: ");
    scanf("%d", &n);
    printf("Enter Block Numbers:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &blocks[i]);
    }
    printf("\nIndex Block : %d\n", indexBlock);
    printf("Allocated Blocks : ");
    for(i = 0; i < n; i++)
    {
        printf("%d ", blocks[i]);
    }
    printf("\n");
    return 0;
}
```

OUTPUT :

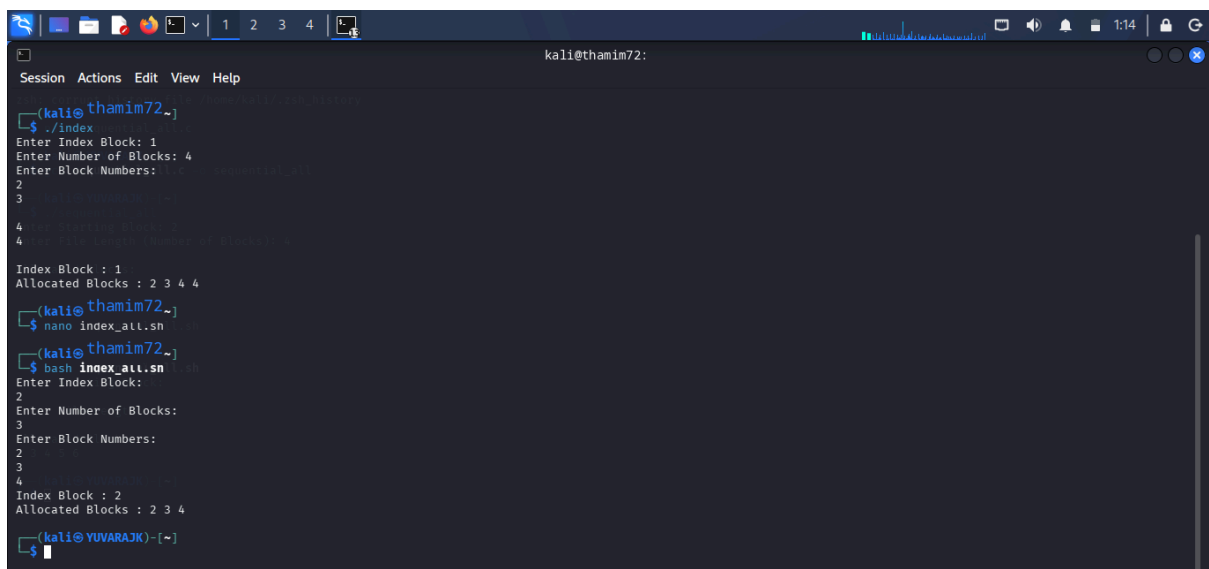
```
kali@thamim72:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@thamim72:~$ nano index_all.c
kali@thamim72:~$ gcc index_all.c
kali@thamim72:~$ gcc index.c -o index
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot find index.c: No such file or directory
collect2: error: ld returned 1 exit status
kali@thamim72:~$ gcc index_all.c -o index
kali@thamim72:~$ ./index
Enter Index Block: 1
Enter Number of Blocks: 4
Enter Block Numbers: 2 3 4 4
2
3
4
4
Index Block : 1
Allocated Blocks : 2 3 4 4
kali@YUVARAJK:~$
```

SHELL PROGRAMMING : INDEX ALLOCATION



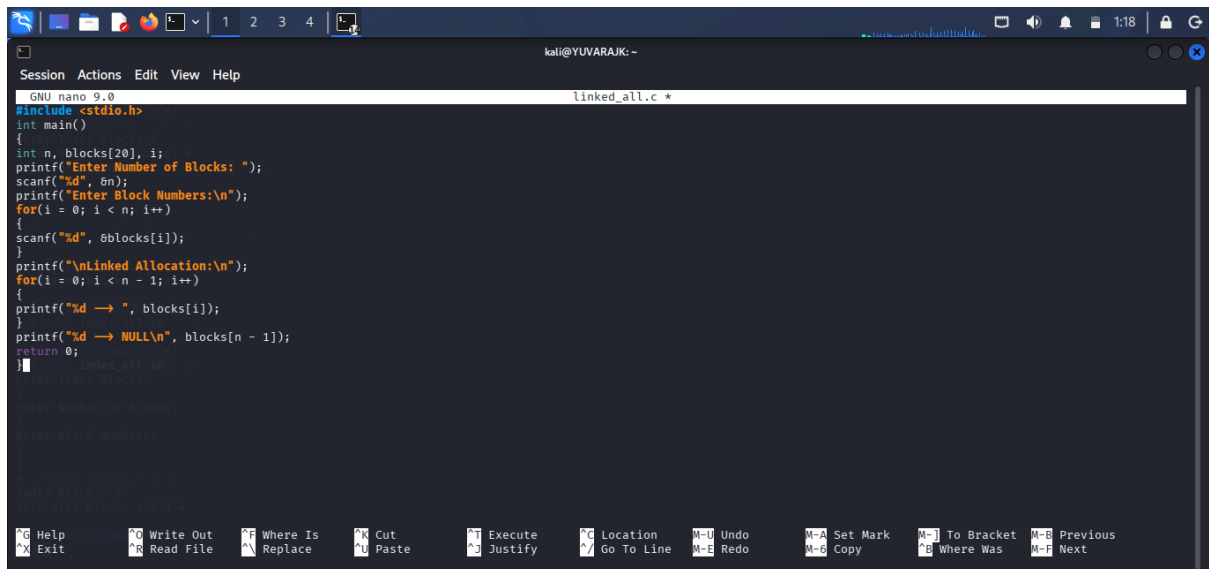
```
GNU nano 9.0 index_all.sh *
#!/bin/bash
echo "Enter Index Block:"
read index
echo "Enter Number of Blocks:"
read n
echo "Enter Block Numbers:"
for ((i=0;i<n;i++))
do
read block[$i]
done
echo "Index Block : $index"
echo -n "Allocated Blocks : "
for ((i=0;i<n;i++))
do
echo -n "${block[$i]} "
done
echo
```

OUTPUT :



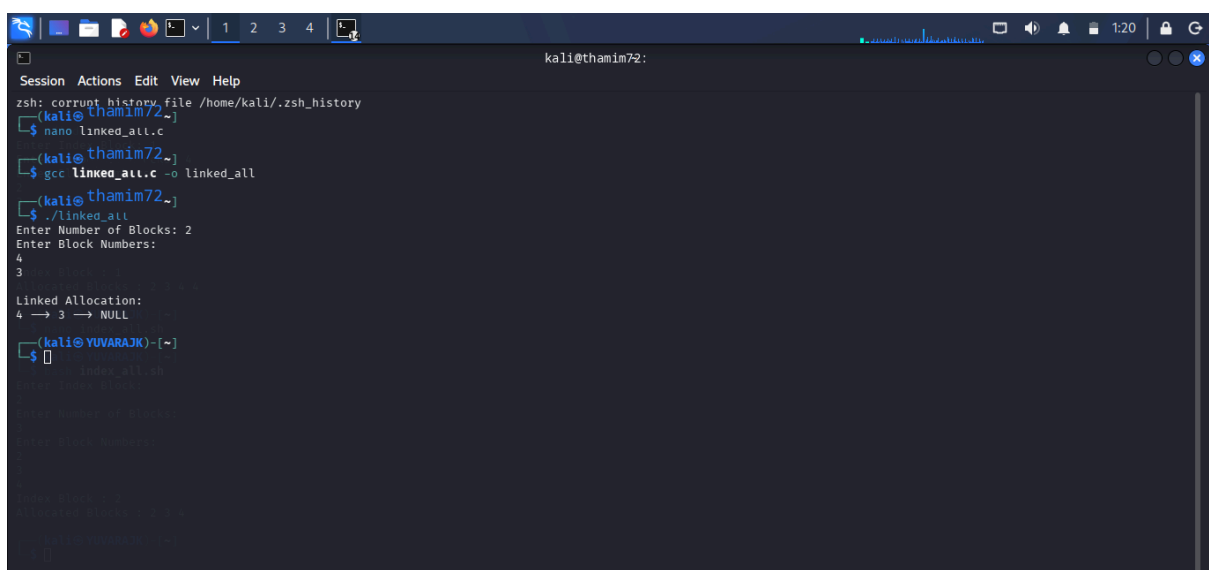
```
(kali@thamim72:~)
$ ./index
Enter Index Block: 1
Enter Number of Blocks: 4
Enter Block Numbers: 1 2 3 4
2
3
4
Index Block : 1
Allocated Blocks : 2 3 4 4
(kali@thamim72:~)
$ nano index_all.sh
(kali@thamim72:~)
$ bash index_all.sh
Enter Index Block: 2
Enter Number of Blocks: 3
Enter Block Numbers: 2 3 4
Index Block : 2
Allocated Blocks : 2 3 4
(kali@YUVARAJK:~)
$
```

C PROGRAMMING : LINKED_ALLOCATION



```
GNU nano 9.0 linked_all.c *
#include <stdio.h>
int main()
{
    int n, blocks[20], i;
    printf("Enter Number of Blocks: ");
    scanf("%d", &n);
    printf("Enter Block Numbers:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &blocks[i]);
    }
    printf("\nLinked Allocation:\n");
    for(i = 0; i < n - 1; i++)
    {
        printf("%d -> ", blocks[i]);
    }
    printf("%d -> NULL\n", blocks[n - 1]);
    return 0;
}
```

OUTPUT :



```
kali@thamim72:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@thamim72:~$ nano linked_all.c
kali@thamim72:~$ gcc linked_all.c -o linked_all
kali@thamim72:~$ ./linked_all
Enter Number of Blocks: 2
Enter Block Numbers:
4
3
Linked Allocation:
4 -> 3 -> NULL
kali@thamim72:~$
```

SHELL PROGRAMMING : LINKED ALLOCATION

[illegible]

OUTPUT :

```

kali@thamim72:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@thamim72:~$ nano linked_all.c
kali@thamim72:~$ gcc linked_all.c -o linked_all
kali@thamim72:~$ ./linked_all
Enter Number of Blocks: 2
Enter Block Numbers:
4
3
Linked Allocation:
4 -> 3 -> NULL
kali@thamim72:~$ nano linked_all.sh
kali@thamim72:~$ bash linked_all.sh
Enter Number of Blocks:
2
Enter Block Numbers:
3
4
Linked Allocation:
3 -> 4 -> NULL
kali@YUVARAJK:~$

```